

IB MATH AA SL

Lesson 5

Exponents & Laws of Indices



Number & Algebra · 60-minute lesson

Syllabus SL 1.5 · Property of Lynda Badmus Education

Do Now — recall

ON YOUR WHITEBOARD

1. Work out $2^3 \times 2^2$

2. Simplify $x^5 \div x^2$

3. What is 5^0 ?

Answers (click to reveal)

1. $2^5 = 32$

2. x^3

3. 1

What we're learning today

LEARNING OBJECTIVES

- 1 Recall**
the laws of indices.
- 2 Apply**
the laws to simplify expressions.
- 3 Interpret**
negative and fractional indices.
- 4 Evaluate**
powers without a calculator.

SUCCESS CRITERIA — I CAN...

- ✓ multiply and divide powers of the same base
- ✓ raise a power to a power
- ✓ rewrite negative indices as reciprocals
- ✓ evaluate fractional indices using roots

The index laws

THE LAWS

$$a^m a^n = a^{m+n} \quad \frac{a^m}{a^n} = a^{m-n} \quad (a^m)^n = a^{mn}$$

Same base: add to multiply, subtract to divide.

Power of a power: multiply the indices.

SPECIAL CASES

$$a^0 = 1$$

anything ($\neq 0$) to the power 0

$$a^{-n} = 1/a^n$$

negative $\rightarrow$ reciprocal

$$a^{(1/n)} = \sqrt[n]{a}$$

fractional $\rightarrow$ root

Simplify

QUESTION

Simplify $(2x^3)^2 \times x^{-4}$,
giving your answer in the form ax^n .

SOLUTION

Step 1 deal with the bracket — square each part:

$$(2x^3)^2 = 4x^6$$

Step 2 multiply by x^{-4} (add indices):

$$4x^6 \times x^{-4} = 4x^{6-4}$$

Step 3 simplify:

$$= 4x^2$$

Your turn — laws

ANSWER IN YOUR BOOK

1. Simplify $x^7 \times x^3$
2. Simplify $(3x^2)^3$
3. Simplify $12x^5 \div 4x^2$

Answers (click to reveal)

1. x^{10}
2. $27x^6$
3. $3x^3$

Negative & fractional indices

REWRITING POWERS

$$a^{-n} = \frac{1}{a^n} \quad a^{\frac{m}{n}} = (\sqrt[n]{a})^m$$

Negative

flip it: $a^{-n} = 1/a^n$

Fractional

denominator = root, numerator = power

EVALUATING

$$\begin{aligned} 8^{(2/3)} \\ = (\sqrt[3]{8})^2 = 2^2 = 4 \end{aligned}$$

$$\begin{aligned} 16^{(-1/2)} \\ = 1/\sqrt{16} = 1/4 \end{aligned}$$

Order

do the root first, then the power.

Evaluate

QUESTION

Without a calculator, evaluate $27^{(2/3)}$.

SOLUTION

Step 1 denominator 3 → cube root:

$$27^{\frac{2}{3}} = (\sqrt[3]{27})^2$$

Step 2 $\sqrt[3]{27} = 3$:

$$= 3^2$$

Step 3 evaluate:

$$= 9$$

Exam-style question

- (a) Simplify $(4x^4y^2)^{1/2}$. [2]
- (b) Evaluate $8^{-2/3}$. [2]
- (c) Write $1/x^3$ using a negative index. [1]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $4^{1/2} \cdot x^2 \cdot y = 2x^2y$ (M1)(A1)
- (b) $8^{-2/3} = 1/8^{2/3} = 1/4$ (M1)(A1)
- (c) x^{-3} (A1)

Common mistakes



Add vs multiply

Multiplying powers ADDS indices; $(a^m)^n$ MULTIPLIES them.



$a^0 = 1$

Not 0 — anything to the power 0 is 1.



Negative $\neq$ negative number

$a^{-2} = 1/a^2$, not $-a^2$.



Fraction order

For $a^{(m/n)}$ take the n th root, then the m th power.

Mixed practice

QUESTIONS

1. Simplify $(2a^3)^4$
2. Simplify $x^6 \div x^{-2}$
3. Evaluate $16^{(3/4)}$
4. Evaluate $25^{(-1/2)}$

Answers (click to reveal)

1. $16a^{12}$
2. x^8
3. $(\sqrt[4]{16})^3 = 2^3 = 8$
4. $1/\sqrt{25} = 1/5$

Plenary & exit ticket

KEY FORMULAS

$$a^m a^n = a^{m+n}$$

$$a^{\frac{m}{n}} = (\sqrt[n]{a})^m$$

EXIT TICKET — before you leave

1. Simplify $y^4 \times y^5$
2. Evaluate $9^{(1/2)}$
3. Write $4^{(-2)}$ as a fraction

Answers (click to reveal)

1. y^9
2. 3
3. $1/16$

Self-assess:
 • need more help
 • nearly there
 • confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Laws of indices.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 6

Logarithms & their laws — the inverse of taking a power.

$$\square_a b = c \quad a^c = b$$